

DISSERTATION TOPIC PROPOSAL

MSc Artificial Intelligence | University of Essex Online

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Proposed Title	Economic Crisis Prediction: Early Warning Systems for Banking Crises. Evaluating the Interpretability–Performance Trade-off Under Structural Non-Stationarity.
Primary Dataset	Jordà-Schularick-Taylor (JST) Macrohistory Database R6
Study Period	1997-2020 (18 advanced economies, 432 country-year observations)
Submission Target	End of June 2026
Supervisor	Dr. Stephania Paladini

1. Research Overview

Economic crises impose severe costs on societies, yet their prediction remains an open challenge. Traditional early-warning systems (EWS) rely on structured macrofinancial indicators, offering interpretability at the cost of missing non-linear signals embedded in textual data. Large language models (LLMs) now make it practical to extract sentiment and uncertainty signals from central bank communications at scale. This dissertation proposes a hybrid framework that combines structured JST macroeconomic features with FinBERT-derived sentiment scores from Bank for International Settlements (BIS) speeches, evaluated against logistic regression and gradient-boosted tree baselines to measure the gain from LLM integration.

2. Study Period and Justification (Question a)

The recommended study window is 1997-2020, giving 24 annual cross-sections across 18 advanced economies (432 country-year observations total).

2.1 Why 1997?

- The BIS central bank speeches corpus — the most academically accessible source of longitudinal central bank text — begins in 1997, making earlier years impractical for sentiment extraction.
- The JST dataset provides excellent data completeness for the core variables (tloans, ltrate, stir, debtgdp) from the mid-1990s onward.
- The 1997 Asian financial crisis provides an early out-of-sample stress event for model validation.

2.2 Why end at 2020?

- JST R6 extends to 2020. The COVID-19 shock in 2020 is structurally distinct (exogenous public health event) from endogenous financial crises — it can be handled via a dummy or robustness sub-sample.
- The Global Financial Crisis of 2008 (19 country-crisis observations in the dataset) provides the primary in-sample stress event and drives substantial statistical power.

Key crisis events within the 1997-2020 window captured in the JST crisisJST indicator:

- Japan banking crisis: 1997
- Global Financial Crisis: 2007-2008 (UK, USA, Belgium, Germany, Denmark, Spain, France, Switzerland, Ireland, Italy, Netherlands, Portugal, Sweden)

This creates an 88-event binary classification problem across the full dataset; approximately 50 post-1997 country-crisis years, providing a meaningful but class-imbalanced training set that requires appropriate treatment (SMOTE or class-weighting).

3. Data Sources (Question b)

3.1 Macroeconomic Panel — Primary (Already Secured)

Source	Key Variables	Access
JST Macrohistory Database R6	59 variables: credit/GDP, LT rate, ST rate, house prices, crisisJST flag	macrohistory.net — Free download (SECURED)
Laeven & Valencia (2020) — Crisis Validation	Systemic banking crisis dates, fiscal costs	IMF Working Paper WP/20/175 — Free PDF

3.2 Sentiment / Text Data — LLM Component (To Be Sourced)

Source	Coverage	URL / Access
BIS Central Bank Speeches Database (PRIMARY)	1997-present. ~18,000+ speeches by central bank governors across all 18 JST countries	bis.org/cbspeeches — Bulk download available (PDF/HTML). Metadata includes date, author, institution.
FOMC Minutes (USA supplement)	1993-present. 8 releases/year. High-quality structured Fed communications.	federalreserve.gov/monetarypolicy/fomc_historical.htm — Free download (optional enrichment for USA).

Note: The BIS speeches database is the recommended primary text source. A Python script using BeautifulSoup or the BIS bulk-download endpoint will collect the text corpus. Speeches are filtered to the 18 JST countries and aggregated annually per country to produce a single annual sentiment score (positive/negative/neutral probability distribution from FinBERT), enabling direct join to the JST panel.

4. Hybrid ML + LLM Model Architecture (Question c)

The proposed framework follows a two-stage feature fusion architecture. This is the most feasible approach within the time budget and produces clearly interpretable results.

4.1 Stage 1: LLM Sentiment Extraction (FinBERT)

- Model: FinBERT (Huang, Wang & Yang, 2023) — a BERT-based model pre-trained on financial corpora and fine-tuned for financial sentiment classification (positive / negative / neutral).
- Input: BIS speeches text, tokenised and passed through FinBERT inference (Hugging Face Transformers library).
- Output per speech: sentiment probability vector [P_pos, P_neg, P_neutral].
- Aggregation: Annual, per-country average sentiment scores, producing a country-year sentiment feature matrix aligned with the JST panel.
- Implementation: Python (transformers, torch). GPU not required for inference on ~18,000 short documents; standard CPU runtime is acceptable.

4.2 Stage 2: ML Early Warning Model

Three model configurations will be trained and compared:

Model	Features	Interpretability	Purpose
Logistic Regression (Baseline)	JST macro only (credit/GDP, rates, house prices)	Native (coefficients)	Interpretable benchmark
XGBoost (M2)	JST macro only	SHAP values	Statistical accuracy benchmark
XGBoost + FinBERT (M3 — Hybrid)	JST macro + annual FinBERT sentiment scores	SHAP values (feature attribution)	Primary hybrid model

4.3 Evaluation Framework

- Metric: AUROC (primary), precision-recall AUC, and Usefulness Score (Alessi & Detken, 2018) to capture the asymmetric cost of missed crises vs false alarms.
- Validation: Expanding-window time-series cross-validation (no data leakage). Train on pre-2000 history; iterate forward annually.
- Interpretability analysis: SHAP beeswarm and waterfall plots to identify which macroeconomic and sentiment features drive predictions in pre-crisis years (e.g., 2006-2007).
- Class imbalance: Handled via `class_weight='balanced'` in sklearn and `scale_pos_weight` in XGBoost; robustness check with SMOTE oversampling.

5. Project Timeline and Realism Assessment (Question d)

The submission deadline of end-June 2026 represents approximately 15 weeks from mid-March 2026. The scope below is deliberately conservative to ensure completion.

Weeks	Phase	Deliverables
1-2	Data Collection & Cleaning	JST panel filtered to 1997-2020. BIS speeches bulk download and parsing. FinBERT environment setup (Hugging Face).
3-4	Sentiment Extraction	FinBERT inference on BIS corpus. Annual country-level sentiment scores. Data join and integrity checks.
5-6	EDA & Feature Engineering	Distribution analysis, crisis class imbalance assessment, feature correlation matrix, lagged variable construction.
7-9	Model Training & Evaluation	Logistic regression baseline, XGBoost macro-only, XGBoost hybrid. AUROC, PR-AUC, Usefulness Score comparison. Expanding-window CV.
10-11	Interpretability Analysis	SHAP plots. Pre-crisis feature attribution analysis. Comparison of M2 vs M3 sentiment contribution.
12-14	Dissertation Writing	Literature review, methodology, results, discussion, conclusion. Target: 15,000-20,000 words.
15	Review, Proofreading & Submission	Final formatting, references, appendices, plagiarism check, submission.

Realism note: The FinBERT sentiment extraction is the most time-sensitive component but is achievable on a standard laptop CPU (estimated ~2-4 hours of inference for the full BIS corpus). The model architecture is deliberately kept to three well-understood variants to avoid scope creep. The June deadline is achievable if data collection is completed by end of March.

6. Research Questions

Main Research Question:

Can the integration of LLM-derived sentiment signals from central bank communications measurably improve the statistical accuracy of ML-based economic crisis prediction models, while preserving sufficient interpretability for risk management applications?

Sub-Questions:

1. Does FinBERT-derived central bank sentiment provide statistically significant predictive power for systemic banking crises over and above standard macrofinancial indicators?
2. What is the trade-off between predictive accuracy (AUROC, PR-AUC) and interpretability (SHAP attribution) when moving from logistic regression to gradient-boosted hybrid models?
3. Which macroeconomic and sentiment features are most predictive in the 1-2 year pre-crisis window, and do they align with the theoretical literature on financial vulnerability?

7. Ethical Considerations

- All data sources (JST, BIS speeches, Laeven & Valencia) are publicly available and carry no personal data considerations.
- The research does not involve human participants and therefore does not require ethics committee review beyond the standard academic ethics declaration.

- Model outputs are retrospective (post-hoc crisis prediction on historical data) and are not intended for live deployment or real financial decision-making.
- LLM use: FinBERT is used as an inference-only tool. No fine-tuning on proprietary or restricted data is performed.

8. References

The following references are limited to the most directly applicable works (Harvard style):

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- Beutel, J., List, S. and von Schweinitz, G. (2019) 'Does machine learning help us predict banking crises?', *Journal of Financial Stability*, 45, article 100693. doi:10.1016/j.jfs.2019.07.005.
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